Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Scalable Catalyst Layer Morphologies and Coating Technologies for PEM and AEM Electrolysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Mechanical Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Chemical Processes	Catalyst Application Technologies	Surface Treatment	Coating / Method Technologies
Concept of Proposed Technology	Polymer Electrolyte Membrane (PEM) and Anion Exchange Membrane (AEM) electrolysis are key technologies for large-scale green hydrogen production. While advances in catalysts and stack designs have been significant, **manufacturing of catalyst layers remains a critical bottleneck**, particularly with respect to scalability, reproducibility, cost, and adaptation to different cell designs. The proposed technology focuses on the **development of scalable catalyst layer morphologies and coating techniques** for PEM and AEM electrolysis, with emphasis on **slot-die coating and spray coating**. The project aims to establish robust structure–process–performance relationships linking coating methodology, catalyst layer morphology, membrane type, and cell architecture. By combining scalable coating technologies, application-oriented catalyst and ink development, and electrochemical validation, the project enables the transition from laboratory-scale electrode fabrication toward **industrial manufacturing of PEM and AEM electrolyzer components**.			

(max 10 pages)

	This approach is particularly relevant for Korean SMEs aiming to develop more competitive and differentiated electrolyzer products, as current manufacturing techniques are often inherited from laboratory or legacy processes and are not always tailored to SME-specific materials, cell concepts, or key performance indicators. The proposed work directly addresses this gap by enabling coating and electrode designs that are optimized for scalability, cost efficiency, and application-driven performance.
Rationale for Joint R&D Project	Korean SMEs play a crucial role in the hydrogen value chain as suppliers of electrolyzer components, materials, and manufacturing equipment. However, access to **manufacturing-relevant electrode technologies** for PEM and AEM electrolysis is limited, particularly for scalable coating and catalyst layer engineering to the specific materials and needs of the SME. This joint R&D project combines: • Fraunhofer UMSICHT's expertise in electrochemical testing, accelerated stress evaluation, and electrode engineering • Korean SMEs' strengths in electrolyzer development, cell design, materials manufacturing, and coating technologies Collaboration is essential to tailor catalyst layers to **industrial-level cell designs of the respective SME**, validate scalability, and reduce technology transfer risks toward commercialization.
Key Words	1. PEM Electrolysis 2. AEM Electrolysis 3. Low PGM / PGM-free catalytic layers 4. Slot-die coating 5.Scalable Manufacturing

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)
R&D Objectives	• Develop scalable catalyst layer morphologies for PEM and AEM electrolysis • Establish slot-die and spray coating as reproducible manufacturing techniques for electrolyzer electrodes • Tailor catalyst layers to different membranes and cell designs • Validate electrochemical performance and durability under accelerated stress conditions			
R&D Target Specifications	• Uniform catalyst layer thickness with deviation ≤ ±10% across coated area • Reduction of noble-metal loading (e.g. Ir for PEM) by ≥20% without performance loss • Stable operation during accelerated stress testing (≥1,000 start–stop or load cycles) • Demonstrated compatibility with multiple membrane and cell designs			
R&D Plan	The joint R&D project is structured into two consecutive phases in accordance with the TIPA International Joint R&D Program: a **6-month pre-study (Phase 1)** followed by a **Main R&D Phase of approximately 2.5 years (Phase 2)**. Phase 1 focuses on rapid feasibility validation and down-selection of coating concepts using accelerated methods, while Phase 2 targets technology maturation, durability, and industrial implementation.			
	Phase 1: Pre-study (6 months) **Objective:** Rapid validation of scalable coating techniques and catalyst layer morphologies for PEM and AEM electrolysis, enabling clear go/no-go decisions for Main R&D.			
	Task 1: Definition of Cell-Specific Requirements and Baseline Benchmarking **Timeline:** Months 1–2 • Definition of technical requirements based on Korean SME partner cell designs (PEM and/or AEM), including operating conditions, compression, flow-field concepts, and target KPIs • Selection of representative membranes, porous transport layers (PTLs), and catalyst systems • Establishment of baseline performance using conventional laboratory-scale coating techniques **Milestones:** • Defined target specifications and KPIs • Baseline electrochemical performance benchmarks			
	Task 2: Catalyst Ink Development for Scalable Coating **Timeline:** Months 1–3 • Formulation of catalyst inks suitable for slot-die and spray coating, focusing on rheology, dispersion stability, solvent systems, and drying behavior • Adaptation of inks to low-iridium PEM catalysts and non-noble or reduced-noble-metal AEM catalysts • Initial assessment of ink robustness under processing-relevant conditions **Milestones:** • Coating-compatible ink formulations • Defined processing windows for scalable deposition			

Task 3: Accelerated Screening of Scalable Coating Techniques
Timeline: Months 2–5

- Fabrication of catalyst layers using slot-die coating and spray coating
- Rapid variation of coating parameters (speed, gap, loading, drying conditions) using accelerated screening approaches
- Morphological characterization (thickness uniformity, porosity, adhesion, ionomer distribution)

Milestones:

- Identification of promising coating strategies
- Quantified reproducibility and scalability indicators

Task 4: Electrochemical Testing and Accelerated Stress Evaluation
Timeline: Months 4–6

- Single-cell PEM and AEM electrolysis testing of coated electrodes
- Accelerated stress tests (load cycling, start–stop, current ramping)
- Correlation of degradation behavior with coating method and catalyst layer morphology

Milestones:

- Performance and durability ranking of coating concepts
- Clear down-selection of coating–material combinations for Phase 2

Outcome of Phase 1:

- Validated feasibility of slot-die and/or spray coating for PEM/AEM catalyst layers
- Down-selected coating strategies tailored to partner-specific cell designs
- Defined risks, optimization potential, and technical scope for Phase 2

Phase 2: Main R&D Phase (~2.5 years)
Objective: Industrial maturation of scalable catalyst layer technologies, durability validation, and preparation for commercialization.

Task 5: Advanced Catalyst Layer Morphology Optimization
Timeline: Months 1–12 (Phase 2)

- Fine-tuning of catalyst layer microstructure (porosity gradients, ionomer distribution, interfacial contact)
- Optimization of noble-metal utilization (Ir reduction for PEM, non-noble strategies for AEM)
- Adaptation of catalyst layers to multiple membrane types and thicknesses

Milestones:

- Optimized catalyst layer designs with improved performance and reduced material cost

Task 6: Scale-Relevant Coating Process Development
Timeline: Months 6–18 (Phase 2)

- Optimization of slot-die and spray coating under scale-relevant conditions (> 300 cm^2 area)
- Assessment of yield, reproducibility, and process robustness in multiple characterisation runs

Milestones:

- Demonstrated scalable coating process with defined quality metrics

Task 7: Cell-Level Integration and Performance Validation
Timeline: Months 12–24 (Phase 2)

- Integration of optimized catalyst layers into Korean SME partner cell designs
- Performance benchmarking against state-of-the-art electrodes
- Validation under realistic operating and load conditions

(max 10 pages)

	Milestones: • Demonstrated cell performance improvements • Verified compatibility with partner-specific cell architectures
	Task 8: Long-Term Durability and Degradation Analysis **Timeline:** Months 18–30 (Phase 2) • Extended durability testing under steady-state and dynamic operation • Identification of dominant degradation mechanisms linked to morphology and coating method • Development of mitigation strategies through materials or process adjustments **Milestones:** • Demonstrated long-term stability • Defined lifetime and reliability indicators
	Task 9: Manufacturability and Techno-Economic Assessment **Timeline:** Months 24–30 (Phase 2) • Evaluation of manufacturing cost drivers and scale-up potential • Preliminary techno-economic assessment comparing scalable vs conventional coating routes • Definition of follow-up pilot-scale or commercialization pathways **Milestones:** • Quantified cost-reduction and scalability potential • Commercialization-ready technical roadmap
	Final Project Outcome The project will deliver **manufacturing-ready catalyst layer and coating technologies** for PEM and AEM electrolysis, enabling Korean SMEs to develop competitive, scalable, and cost-efficient electrolyzer components and strengthening Korea's position in the global hydrogen supply chain.

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer UMSICHT will act as the **technical lead and neutral innovation partner**, providing expertise in electrochemical systems, catalyst layer engineering, and scalable manufacturing processes for PEM and AEM electrolysis tailored to the needs of Korean SMEs **Technical Contributions and Research Scope** • Development and optimization of scalable catalyst layer fabrication concepts for PEM and AEM electrolysis • Design, implementation, and operation of slot-die coating and spray coating processes for electrolyzer electrodes • Catalyst layer morphology engineering, including control of porosity, ionomer distribution, thickness, and interfacial adhesion • Application-oriented catalyst and ink development tailored to scalable coating techniques **Testing, Evaluation, and Infrastructure** • Electrochemical performance testing in single-cell PEM and AEM electrolyzers • Accelerated stress testing (load cycling, start–stop, dynamic operation) to assess durability and degradation behavior • Advanced characterization of catalyst layers and correlation of structure–process–performance relationships **Resources and Project Coordination** • Provision of laboratory infrastructure for coating, electrochemical testing, and materials characterization • Development of benchmarking protocols, quality metrics, and evaluation frameworks relevant for industrial manufacturing • Technical coordination of joint R&D activities and facilitation of knowledge transfer to Korean SME partners
Expected R&R of Korean SME Partner	The Korean SME partner is expected to contribute **industrial know-how, application requirements, and development capabilities** relevant to PEM and/or AEM electrolysis systems. **Technical Contributions and Development Tasks** • Provision of PEM and/or AEM electrolyzer cell designs, operating conditions, and target performance indicators • Contribution to the adaptation of catalyst layers and coating processes to partner-specific cell architectures and manufacturing constraints • Participation in catalyst layer fabrication, coating scale-up, and process optimization activities **Testing, Validation, and Industrial Perspective** • Validation of coated electrodes in SME-specific test platforms or prototype cells • Evaluation of manufacturability, reproducibility, and integration into existing production workflows • Feedback on industrial feasibility, cost targets, and performance requirements **Collaboration and Commercialization Efforts** • Active participation in joint experimental work, data evaluation, and technical decision-making • Contribution to defining scale-up strategies, pilot production concepts, and commercialization pathways • Exploration of follow-up product development or market entry opportunities based on project outcomes Through this collaborative framework, the project ensures that scalable catalyst layer technologies are **technically validated, industrially relevant, and directly transferable** to Korean SMEs for commercialization.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project will deliver **manufacturing-ready catalyst layer and coating technologies** for PEM and AEM electrolysis, addressing one of the key bottlenecks in scaling green hydrogen production. **Technological Outcomes** • Validated **slot-die and spray coating processes** for PEM and AEM catalyst layers with high reproducibility and scalability • Optimized catalyst layer morphologies tailored to different membranes and cell architectures, improving performance and durability • Demonstrated reduction of noble-metal usage (e.g. iridium for PEM) through improved catalyst utilization • Established structure–process–performance relationships enabling predictable electrode design and manufacturing **Commercialization Potential** • Enablement of Korean SMEs to transition from laboratory or legacy electrode fabrication methods toward **industrial-scale manufacturing** • Creation of proprietary electrode designs, coating processes, and manufacturing know-how suitable for commercialization or licensing • Faster development cycles and reduced scale-up risk for PEM and AEM electrolyzer components **Industrial and Economic Impact** • Strengthening of Korea's domestic electrolyzer supply chain by reducing reliance on imported electrode technologies • Potential manufacturing cost reductions of **15–30%** through scalable coating, higher yields, and lower noble-metal loadings • Improved competitiveness of Korean SMEs in global hydrogen markets through differentiated, manufacturable PEM and AEM technologies Overall, the project supports Korea's hydrogen strategy by enabling **cost-efficient, scalable, and locally developed PEM and AEM electrolyzer technologies**, positioning Korean SMEs as key suppliers in the global green hydrogen value chain.

6. References

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(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)